

LESSON 1: INTRODUCING COMPUTERS

Computers are all around us. We use them in almost every aspect of our lives. But what are they? How do we best define them? What do they do for us?

A **COMPUTER** IS A TECHNOLOGY WHOSE PURPOSE IS TO PROCESS INFORMATION ACCORDING TO A STORED PROGRAM.

TECHNOLOGY

The word technology refers to the tools and machines we use to help us perform various tasks. Every technology has a specific purpose; it is designed for a particular task.

EXAMPLE 1.1

A pencil is a form of technology. It is a tool designed to help us perform the task of writing. Even though there are many different types of pencils, they are all designed to perform the same task, writing.



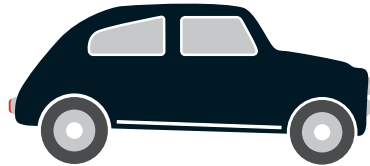
Notes

The term 'technology' encompasses more than simply tools and machines; it includes any practical application of knowledge.

The Merriam-Webster Dictionary defines technology as "the practical application of knowledge, especially in a particular area."

EXAMPLE 1.2

A car is also a form of technology. It is a machine designed to help us travel from one place to another on land. Regardless of the shape, size and color, all cars are designed for the specific task of transporting people from one place to another on land.



EVERY TECHNOLOGY IS DESIGNED FOR A PURPOSE.

PROCESSING

To process something means to change it from one form to another. Processing is not any kind of change; it is a very structured form of change that can be replicated. A process is structured according to a specific set of steps; if given the same starting conditions and materials, these steps will provide the same end result every time they are performed.

EXAMPLE 1.3

Processed food is food that undergoes change starting from certain raw materials. Bread is processed food. It starts as wheat, a raw material that is unprocessed. The processing of wheat to make bread must follow a specific set of steps.

A PROCESS IS A SERIES OF STEPS THAT COMPLETE A PARTICULAR TASK AND CAN BE REPLICATED.

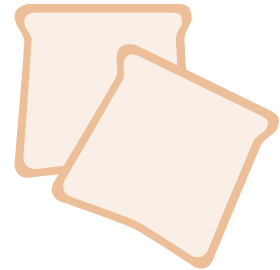
Anyone who replicates these steps will make bread from wheat. Regardless of how many times the steps are repeated and regardless of who performs these steps, the processing of wheat will always result in bread.



PROCESSING OF WHEAT TO BREAD

STEPS

1. *Clean the wheat grains*
2. *Grind the wheat grains*
3. *Sift the flour to remove the husk*
4. *Send the flour to the bakery*
5. *Use flour to make bread dough*
6. *Bake the dough*



INFORMATION

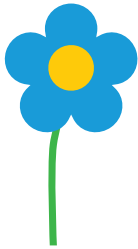
In general, the term information refers to anything meaningful that tells us something about ourselves or the world around us. This book contains information. The information is conveyed in the words and pictures of this book.

As humans, we process information. We see, feel and hear things that we process to gain information about the world around us. We retain this information in our memories. Later, we may use this information to interact with the world.

EXAMPLE 1.4

Sarah and her friends were playing the “Guess the Object” game. Sarah was blindfolded, and her friends laid out objects in front of her. Using her senses, she had to gather information about these objects to guess what they were.

Even without seeing the objects, Sarah could identify them. In addition to each object's shape, which she could feel, the flowers smelled nice, the keys jingled and the ice cube felt cold.



Sarah's senses helped her gather information about each object. But that was not enough to identify each object. Sarah must have had in her memory some information about the characteristics of each object against which she could compare. The new information gathered by her senses was compared with the information she had in her memory.



[1]

Sarah gets information by smelling the object.



[2]

Sarah processes the information by comparing it with previous information in her memory.

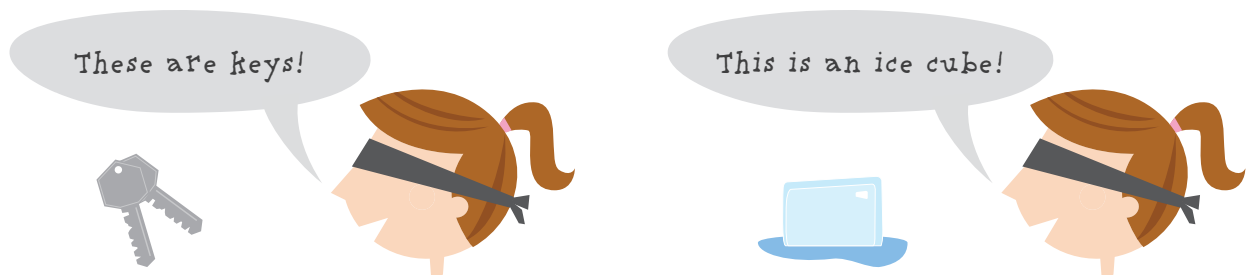
To identify the object, Sarah's brain had to process information. Sarah's brain had to go through a series of steps in which she compared the information acquired through her senses with the information she had in her memory. By making these comparisons, Sarah was able to identify the object and state its name. The information acquired through her senses was processed; it was changed to the name of the object.

[3]

Sarah informs her friends that the object is a flower.



Sarah followed the same steps to identify the rest of the objects. Her senses acquired information about each object. She compared that information with that in her memory. An object was identified when a match was found. She would then inform her friends of the name of each object.



[4]

Sarah follows the same steps to identify the other two objects.

COMPUTERS

Computers are a form of technology designed for the purpose of processing information. This includes creating information, sharing information and storing information, all of which require some form of processing.

CREATING INFORMATION

People use computers to create new information. When we type a text message, email or essay, in each case, we are creating new information. When we record a voice message or video, we are creating new information. In this context, information is considered to be any meaningful content. Making new content, whether in the form of text, audio or images, is all made possible by computers.

Creating new information is not limited to typing an essay, recording a voice message or capturing an image. Creating new information also includes the processing of data either through calculation or analysis.

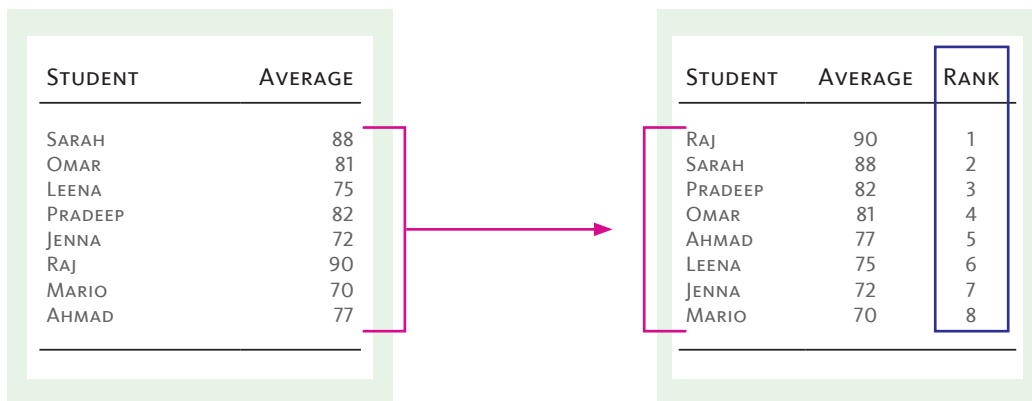
EXAMPLE 1.5

Every school keeps track of its students' grades. At the end of each term, the school issues a report card to every student. The computer helps create every report card. Teachers enter the grades for each subject into the computer system. The computer then processes this information to provide an average for each student. The average is the new information.

SUBJECT	GRADE
MATH	75
ENGLISH	78
SCIENCE	82
SOCIAL STUDIES	74
IT STUDIES	80
ART	83
MUSIC	70
PE	82
AVERAGE	78

Data is processed to calculate the average.

The school ranks its students. To do so, the list of students needs to be sorted from the highest average to the lowest. The computer processed this information and sorted the list of students. This provides the school with the student ranks, which is new information.



Data is processed to sort the list of students from highest to lowest average. The rank is new information.

SHARING INFORMATION

Creating information and new content becomes more valuable when it can be shared with others. Computers have tremendously increased our ability to share information.

Sharing information is made possible by the Internet, a complex network of telecommunication technologies that spans the globe and allows computers to connect to each other and to other digital devices.

THE INTERNET IS A COMPLEX NETWORK OF TELECOMMUNICATIONS TECHNOLOGIES THAT CONNECTS COMPUTERS ACROSS THE WORLD.

All information accessible through the Internet is referred to as the World Wide Web or simply as the Web.

Information on the Web is organized into files. The content of each file can be text, audio or images. This content is displayed as webpages, music, pictures or video. Each file has a specific address, a web address, that is used to locate it.

AN ADDRESS IS USED TO LOCATE INFORMATION.

Information on the Web is shared as either private or public, which determines who gains access to this information. Access to private information is restricted and requires a password. In contrast, public information does not require a password and is accessible to everyone.

EXAMPLE 1.6

Your school may have a website. The school website is made up of several web pages that contain information made public by your school. In addition, the school may offer its students a portal through which they can sign in with their passwords to access their private information, such as their grades.

USERNAME:

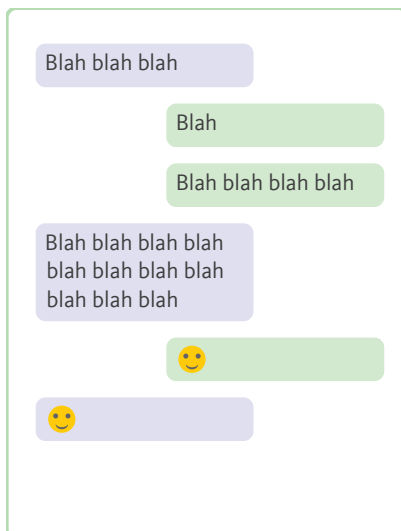
PASSWORD:

COMMUNICATION IS A FORM OF SHARING INFORMATION THAT RESULTS IN AN EXCHANGE BETWEEN TWO OR MORE PEOPLE.

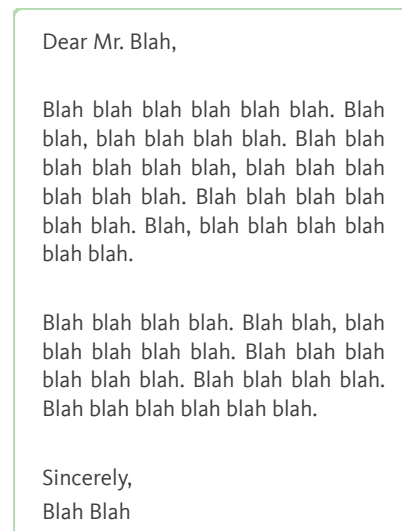
Another form of sharing information is communication. Communication is sharing that results in an exchange of information between two or more people.

Two methods of communication facilitated by computers are messaging and emailing. Both methods allow for the exchange of information in the form of text and audiovisual content. Messaging is better suited for shorter and more frequent exchanges of messages, whereas emailing is better suited for more prolonged and less frequent exchanges of messages.

Text messages



Email



Messaging and emailing are two forms of communication facilitated by computers.

STORING INFORMATION

Once information has been created, it must be stored. Storage of information is important. If information were volatile, meaning it “evaporates” whenever a computer was powered off, no information would be left. All our information would be lost; every message, every essay and every photo would be lost.

Every computer must be capable of storing information. Storage provides a permanent repository for information. Computers not only allow us to store our information, but they also provide us with the facilities to search and find specific items of information kept in storage.

STORAGE IS THE PERMANENT REPOSITORY FOR INFORMATION.

All information is stored in the form of files. Every file is given a filename by which it can be identified. Every file is also given a specific address at which it can be located. The size of a file is highly variable; it can contain 1 byte or several gigabytes of information. The content of a file may be mostly instructions, which is the case for apps, or it may only be data, as is the case for photos.

A FILE IS A UNIT OF STORABLE INFORMATION.

THE STORED PROGRAM

The stored program concept is an important concept that made the modern computer possible. Imagine your computer without any apps. What could it

possibly do? Would it be of any use to you?

The fact is that every app on your computer performs a particular task. If you want to compose an essay, you use a word processing app. If you want to record a video, you use a camera app. If you want to search for information on the Web, you use a web browser app. Each of these apps is a program stored in your computer, and each program contains the instructions the computer must follow to perform that particular task.

A PROGRAM IS A SET OF INSTRUCTIONS INTENDED TO PERFORM A PARTICULAR TASK.

In addition to the apps you use, there are other stored programs on your computer that you do not see and do not directly use, but they are critical for making your computer work the way it does. Most of these programs are part of the operating system, which is a group of programs that work together to allow your computer to use a variety of apps and to function properly.

SUMMARY

The word technology refers to the tools and machines we use to help us perform various tasks. Every technology is designed for a purpose.

Processing is a structured form of change that can be replicated. Computers are designed for the purpose of processing information.

Computers allow us to create, share and store information.

The Internet is a complex network of telecommunications technologies that connects computers across the world. It is used to share and exchange information. Access to shared information is either restricted, private or unrestricted, public.

An address is used to locate information. A web address is an address used to locate information on the World Wide Web.

Communication is a form of sharing information that results in an exchange between two or more people. Emailing and messaging are two examples of methods of communication.

Storage is the permanent repository for information. All information is stored as files. A file is a unit of storable information. Every file has a filename by which it is identified and an address by which it is located.

A program is a set of instructions intended to perform a particular task. Stored programs allow the computer to perform different tasks. An app is an example of a stored program.

A computer is a technology whose purpose is to process information according to a stored program.

Activity 1.1

Read the following statements, and write on the line provided for each statement 'T' if it is true and 'F' if it is false.

T/F	STATEMENT
<u>T</u>	Information is any meaningful content.
<u>F</u>	Computers are not a form of technology.
<u>T</u>	Every technology has a purpose.
<u>T</u>	Processing generally means changing something in accordance with a series of steps.
<u>F</u>	Information cannot be processed.
<u>F</u>	Computers allow us to only share information.
<u>T</u>	Storage is the permanent repository for information.
<u>T</u>	A program is a set of instructions intended to perform a particular task.
<u>F</u>	An application is not an example of a stored program.
<u>T</u>	Stored programs made modern day computers possible.

Activity 1.2

Identify machines, appliances and devices for each of the forms of technology based on its stated purpose.

1. Which two of these machines are forms of technology designed for the purpose of transportation?
 - a) ship
 - b) microwave oven
 - c) airplane
 - d) computer

2. Which two of these appliances are forms of technology designed for the purpose of cooling food?
 - a) washing machine
 - b) freezer
 - c) air conditioner
 - d) refrigerator

3. Which two of these devices are forms of technology designed for the purpose of generating electrical energy?
 - a) wind turbine
 - b) phone battery
 - c) solar panel
 - d) electric motor

4. Which two of these devices are forms of technology designed for the purpose of processing information?
 - a) laptop
 - b) television
 - c) printer
 - d) desktop

Activity 1.3

Answer the questions below.

1. Transportation technologies can be further categorized based on the medium within which they are intended to travel. List two machines designed for the purpose of traveling in each medium.

Land : car, train, bus

Air : airplane, helicopter, hot-air balloon

Water : ship, submarine, yacht

2. Investigate the two forms of technology. Identify and state the purpose of each.

Nanotechnology : its purpose is to manipulate matter at the tiniest scale, the nanoscale (1 -100 nm), to make new products.

Biotechnology : its purpose is to use biological processes, living cells and organisms to make new products.

3. “Recognizing the purpose of any particular form of technology is important.” Analyze this statement, investigate and provide at least two supporting reasons.

Recognizing the purpose behind a particular technology allows us to:

1) understand how to use it.

2) determine if it will be useful to us.

3) avoid potential risks.

Activity 1.4

Answer the questions below.

1. What information do drivers get when they see a red traffic light?
 - a) go
 - b) **stop**
 - c) turn left
 - d) turn right

2. What information do drivers get when they hear the sirens of an ambulance behind them?
 - a) **move out of the way**
 - b) block the road
 - c) get help
 - d) cover your ears

3. If you were blindfolded and asked to guess if an object in front of you was an ice cube, what information would you be searching for?
 - a) **cold, wet, smooth**
 - b) warm, dry, rough
 - c) cold, wet, rough
 - d) warm, wet, smooth

4. For each of the questions above, state the sense through which information was acquired.

Question 1: **sight**

Question 2: **hearing**

Question 3: **touch**

Activity 1.5

Answer the questions below.

1. Which of the following best describes a process?
 - a) A random sequence of events
 - b) A type of machine used in manufacturing
 - c) A clear sequence of steps for completing a task
 - d) A finished product

2. Which of the following is a characteristic of a well-defined process?
 - a) It is constantly changing
 - b) It is repeatable and consistent
 - c) It does not have a clear beginning
 - d) It does not have clear steps to follow

3. Which of the following is not an example of a process?
 - a) Baking a cake
 - b) Getting dressed
 - c) Watching a movie
 - d) Brushing teeth

4. Why do you think it is important to understand a process?
 - a) It helps us complete the task quickly and properly
 - b) It allows us to skip steps in the process
 - c) It makes the process more complicated than it needs to be
 - d) It ensures that you don't need to follow any specific steps

Activity 1.6

Answer the questions below.

1. What information do you need to bake a cake?
 - a) Which ingredients go into making a cake
 - b) What steps need to be performed
 - c) Which utensils are required to perform the steps
 - d) All the above

2. What item of information would be useful for you to navigate your way out of your classroom?
 - a) The color of the walls
 - b) The location of the exit
 - c) The names of the students
 - d) The number of open windows

3. “Information is critical to the process of decision-making.” Is this statement true? Why? Give an example.

Yes, this statement is true. Having the correct information allows us to make good decisions and to avoid making mistakes. The lack of information or having the wrong information can lead to unwanted outcomes. For example, when going on a road trip, having the wrong location for the destination or not having the location may result in us getting lost and not arriving at the destination.

Activity 1.7

Match each word with the statement that best describes it.

	File	Filename	Program	Address
	Web Address	Internet	Storage	
1.	A telecommunications technology that spans the globe	Internet		
2.	A unit of storable information	File		
3.	It is used to locate a file	Address		
4.	It is used to identify a file	Filename		
5.	It is used to locate a file across the Internet	Web Address		
6.	A computer's permanent repository of information	Storage		
7.	A set of instructions intended to perform a task	Program		

Activity 1.8

For each of the following scenarios, identify how information was created, stored and shared.

- Karl watches a video on YouTube.

Created : The video is the information created by someone.

Stored : The video is stored on a YouTube computer (server).

Shared : The video is viewed by people who find it , like Karl.

2. Jenna uses her smartphone to send a voice message to her friend.

Created : The voice message is the information created.

Stored : The voice message is stored on Jenna's smartphone.

Shared : The voice message is sent through the Internet to her
friend's phone.

3. A teacher enters her students' grades. Later, parents view their children's report cards which contain their general average and rank.

Created : 1) Grades are entered. 2) General averages are calculated.
3) Ranks are identified by sorting based on general averages.

Stored : Student grades, general averages and ranks are stored in
the school's main computer (server).

Shared : Parents access the school's main computer using the Internet
and view their child's grades, general average and rank.

4. Describe your own example of using a computer to create, store and share information.

Activity 1.9

Match each word with the statement that best describes it.

	Program	Private	Messaging	Public
	Email	Applications	Images	
1.	Frequent and short exchanges of information		<u>Messaging</u>	
2.	Information that has restricted access		<u>Private</u>	
3.	Information that has unrestricted access		<u>Public</u>	
4.	Communication suitable for long, infrequent exchanges		<u>Email</u>	
5.	Examples of files that contain only data		<u>Images</u>	
6.	Examples of files that contain both instructions and data		<u>Applications</u>	
7.	A set of instructions that perform a particular task		<u>Program</u>	

Activity 1.10

Answer the questions below.

- Two common methods of communication are messaging and email. Describe how the two methods differ.

Messaging may be more effective for quick and informal communication with
friends and classmates, while email may be more effective for more formal or
professional communication with teachers or business partners.

2. In the previous question, the differences between messaging and email were described. Describe how the two methods are similar.

1) Both are forms of digital communication.

2) Both require the Internet to transfer information.

3) Both can be used to send information in the form of text, audio and video.

3. Every application performs a particular task. List two applications and what you use them for.

Answers may vary. For example, I use PowerPoint to prepare a presentation

for my classmates about my science project.

4. List at least three ways you will use computers to help you learn.

Possible answers may include reference to: 1) access to educational

resources, 2) collaboration and communication, 3) interactive learning,

4) personalized learning and 5) research and data analysis.
